

## Scalability & Future Plan

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 15 July 2026                                                |
| <b>Team ID</b>       |                                                             |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 1 Mark                                                      |

### Scalability & Future Plan:

This document outlines how the OptiCrop solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                       | Impact                                                                 | Priority to Address |
|------|--------------------------------------------------|------------------------------------------------------------------------|---------------------|
| 1    | Local deployment only — not cloud hosted         | App is only accessible on local machine — cannot be shared publicly    | High                |
| 2    | Limited to single ML model (Logistic Regression) | May not perform optimally for all crop types and regional conditions   | Medium              |
| 3    | No real-time weather API integration             | Farmers must manually input climate data instead of automatic fetching | High                |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                         | Proposed Upgrade / Solution                                                                             |
|------|--------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Single user — local Flask dev server  | Deploy on cloud platform (AWS/Heroku/Render) with load balancing for multiple concurrent users          |
| 2    | Data Storage       | CSV file with 2200 records locally    | Migrate to PostgreSQL/MongoDB database for larger datasets, real-time updates, and user history storage |
| 3    | Performance        | Single ML model — Logistic Regression | Implement ensemble models (Random Forest, XGBoost) and AutoML for improved prediction accuracy          |

|   |          |                                 |                                                                                                     |
|---|----------|---------------------------------|-----------------------------------------------------------------------------------------------------|
| 4 | Security | No authentication — open access | Add user authentication (JWT/OAuth), input sanitization, and HTTPS for secure production deployment |
|---|----------|---------------------------------|-----------------------------------------------------------------------------------------------------|

### Future Roadmap:

| Phase   | Planned Feature / Enhancement                                                                    | Target Timeline | Expected Impact                                                                |
|---------|--------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------|
| Phase 2 | Cloud Deployment on Heroku/Render + Real-time Weather API integration for automated climate data | 3–6 months      | Public accessibility for farmers across India; eliminates manual climate input |

|         |                                                                                                                              |              |                                                                                               |
|---------|------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------|
| Phase 3 | Mobile App (Android/iOS) + Regional Language Support (Telugu, Hindi) + Advanced ML models (Random Forest, XGBoost)           | 6–12 months  | Wider farmer adoption; improved prediction accuracy across diverse agricultural regions       |
| Phase 4 | IoT Sensor Integration for automated soil parameter detection + AI-powered crop disease detection using image classification | 12–18 months | Fully automated smart farming system; real-time crop health monitoring and disease prevention |